

Metric Bone-Surface Reconstruction in Confined Surgical Corridors with a Near-Coaxial Photometric-Stereo Endoscope

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Abstract—Surgery on bone increasingly happens at the end of a narrow access corridor—a tubular retractor, an arthroscopic portal, an endonasal approach—where navigation needs the exposed surface in metric coordinates but the corridor admits no stereo baseline, projector, or tracked pointer. The endoscope already inside could supply that surface, except that a monocular scope has no intrinsic length reference beyond its own illumination geometry. We present a near-field photometric-stereo method for an 18 mm endoscope carrying four LEDs on a ring of radius 6.05 mm, and show that the ring baseline suffices to place the surface metrically from a single pose. Four lights overdetermine the three per-pixel unknowns, so the residual of a solve whose normals are tied to a candidate surface carries geometric information: minimised over a two-parameter depth family—working distance and relief gain—it selects the surface placement, with absolute scale entering through the measured ring radius and the lens depth-of-field band as prior. Validated on a CT-instrumented lumbar spine phantom—a demanding corridor target, with per-vertebra rigid-body structure—the method reconstructs all three well-conditioned vertebral levels to 2.7 mm to 3.1 mm fiducial RMS under a rigid 6-DOF alignment, with per-level scale errors of 10% to 21%. Under the same protocol, classical far-field photometric stereo suffers a systematic low-frequency bowl (6.6 mm on its worst well-conditioned level) and inverse-square shading-to-range fails on every level (6.4 mm to 9.6 mm); a variant given oracle depth anchoring bounds the achievable accuracy at 1.3 mm to 1.7 mm. Unmodelled as-built LED geometry biases the recovered working distance by up to 15 mm and is the dominant remaining error; the two failing levels are traced to a three-fiducial degeneracy and a confidence-mask collapse.

I. INTRODUCTION

BONE-directed surgery is migrating into narrow access corridors: tubular retractors in the spine [1], arthroscopic portals in joints, endoscopic approaches to the skull base. Navigation against preoperative imaging then hinges on registering the exposed bone surface—in the spine, per vertebra, since the column flexes at the discs [2]. Whether guidance is manual or robotic [3], that registration currently costs either intraoperative 3-D imaging (cone-beam CT / O-arm, with added radiation and workflow interruption [4]) or tracked instruments that demand line of sight into a corridor that barely admits the scope. Yet the endoscope is already looking at the bone; if it could measure that surface *metrically* on its own, the surface itself could drive registration with no added radiation or tracking hardware.

Metric shape from the scope already in the wound is, however, exactly where existing pipelines fall short [5]. Multi-view geometry—structure-from-motion, SLAM, and their learned variants [6], engineered for intra-articular reconstruction in Dense-ArthroSLAM [7]—needs texture and parallax that wet,

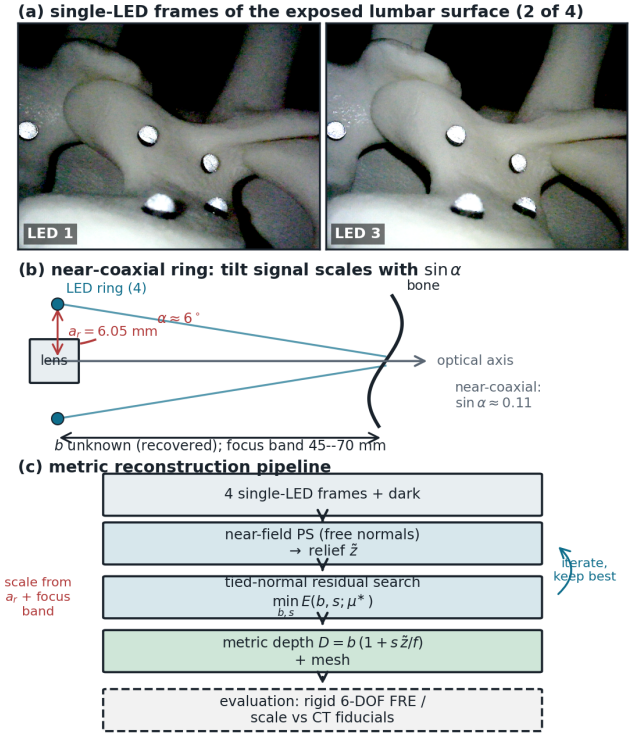


Fig. 1. **System overview.** (a) Two of the four single-LED frames of an exposed lumbar segment; the shading change between opposing LEDs encodes surface tilt, and the specular dots are the evaluation fiducials. (b) The LEDs sit on a ring of radius $a_r = 6.05 \text{ mm}$, only $\alpha \approx 6^\circ$ off-axis at the working distance: near-field and near-coaxial. (c) Pipeline: a free-normal near-field solve proposes relief $\tilde{z}$; a tied-normal residual search recovers working distance b and relief gain s , with metric scale entering through a_r ; the reconstruction is scored against CT fiducials (dashed).

low-texture bone provides poorly, and monocular reconstruction is in any case scale-blind without an external reference. Photometric stereo [8] fits the setting better: it recovers a dense normal field from a fixed viewpoint under varying illumination and is albedo-invariant, which matters on stained, textured bone. It has been demonstrated through endoscopes on soft tissue [9]–[11], but with accuracy unreported, mapped only per pixel, or confined to a bench target (Sec. II); none is evaluated against tomographic ground truth. Near-field formulations with per-pixel light directions and inverse-square attenuation are well understood [12]–[14], yet they obtain absolute scale from outside the images—a known or tracked working distance, or a calibrated auxiliary sensor. The scale, in principle, is

already in the illumination: the decline of the scope’s own light encodes range, and recent work recovers metric depth from it [15]–[17]. But that line has been developed for single co-located sources with light–camera baselines of ~ 3 mm or smaller, on deformable gastrointestinal tissue, and its accuracy has not been tested against the millimetre demands of surgical navigation on rigid anatomy.

The obstacle specific to a small bone-facing scope is geometric. A ring of LEDs around the objective is *near-coaxial*: at $b \approx 55$ mm each light is only $\alpha \approx 6^\circ$ off-axis, so the tilt-encoding lateral component of every light vector scales with $\sin \alpha \approx 0.11$, the conditioning of the four-light system degrades as $\sqrt{2} \cot \alpha$, and a far-field model integrates its misfit into a low-frequency bowl that no global scale or offset can remove. Errors in the assumed light geometry are amplified by the same factor, so the wiring, aim, and emission profile of the LEDs must be identified rather than assumed.

This paper presents a metric near-field photometric-stereo pipeline for that regime (Fig. 1). A free-normal near-field solve proposes a relative relief map; the reconstruction then searches a two-parameter metric depth family—working distance b and relief gain s —by minimising the four-image photometric residual of the candidate surface with normals *tied* to it, so that a wrong geometry produces a shading misfit the per-pixel solve cannot absorb. Because four lights overdetermine the three per-pixel unknowns—the classical virtue of four-source photometric stereo [18], [19]—this residual is informative, and because the ring radius a_r is a measured hardware length, its minimum selects an absolute scale; the only prior is the lens depth-of-field band, a bench property of the fixed-focus optics. The rig itself is identified in situ: the LED wiring by a permutation study against the fiducial depths of two levels, the beam aim by a centroid analysis, and the emission exponent by an end-to-end sensitivity study.

The contributions are:

- a metric formulation of near-field photometric stereo for near-coaxial ring illumination: a tied-normal residual objective over a two-parameter depth family whose absolute scale comes from the LED ring baseline, optimised by a multi-start, keep-best search inside a bench-measured focus band (Sec. IV);
- in-situ identification of a ring-lit endoscope: index→azimuth wiring by a 24-permutation study ($r = +0.97/+0.99$ vs. $-0.97/-0.99$ for the diametric flip), a beam-centroid analysis that separates LED *aim* from LED *position* and reveals convergent aim on this rig, and a sensitivity study showing that the photometric residual under-constrains the emission exponent while final accuracy depends on it strongly (Secs. III, IV-E);
- a fiducial-based evaluation on a CT-instrumented lumbar phantom across five vertebral levels—rigid-only 6-DOF FRE of 2.7 mm to 3.1 mm on the three well-conditioned levels, registration-free scale checks, ablations against far-field, inverse-square, and oracle-anchored variants, and an analysis of both failure cases (Sec. V).

II. RELATED WORK

A. Photometric Stereo and Near-Field Illumination

Woodham’s formulation [8] recovers per-pixel normals from ≥ 3 distant illuminations, integrated to depth by, classically, the Frankot–Chellappa projection [20]; four sources add the redundancy needed to reject highlights and shadows [18], [19]. Non-Lambertian extensions (robust, parametric, example-based, learned) attack the reflectance term but retain far-field lights [21], under which shape is recoverable only up to the generalised bas-relief ambiguity [22]. Near-field sources change both statements: Quéau et al. [12] give the complete LED forward model—position, principal direction, $\cos^\mu$ anisotropy, inverse-square fall-off—with calibration and solvers, the attenuation term makes near-source shape-from-shading well-posed [23], and the LUCES benchmark quantifies both the necessity and the ceiling of the model: a naive far-field method degrades to 37.5° mean normal error at 10–30 cm, while the best calibrated near-field solvers, given 52 individually calibrated LEDs, average 13.3° normal and 3.2 mm depth error [24]. Semi-calibrated variants estimate light intensities in the solve [25], uncalibrated near-light photometric stereo has been analysed in [26], and dedicated procedures calibrate point-light positions [27]. All of this work estimates normals (or depth up to the modelled lights) with the working distance essentially known; the present paper inverts the emphasis and estimates the metric surface placement itself, using the ring baseline as the length reference.

B. Endoscopic Reconstruction and Metric Scale

That an endoscope’s illumination geometry is special was recognised early: Okatani and Deguchi [28] solved shape-from-shading with the source at the projection centre, the limiting case our 6° -off-axis ring sits just beyond. Photometric stereo endoscopy was introduced by Parot et al. [9] on phantoms and *ex vivo* human gastrointestinal tissue, and engineered into a 14 mm clinical gastroscope used in vivo in eight human rectums [29]—but the technique is, by its authors’ own description, qualitative in height: low-frequency shape is discarded by a high-pass filter, and across the programme no surface-accuracy figure was reported. Close-range colour-multiplexed PS in laparoscopy [10] computes absolute depth from a calibrated attenuating light model, but reports only per-pixel error maps spanning 1 mm to 6 mm *ex vivo* at ~ 35 mm range, with no summary statistic; capsule PS with four LEDs at 5.5 mm offset [11] reaches 0.51 mm RMSE, but at 21 mm working distance (a baseline-to-depth ratio $2.3\times$ ours), on a single smooth convex target, with the metric seed taken from a specular highlight whose millimetre-level error triples the surface error. Batlle et al. [13] photometrically calibrate a clinical colonoscope to three grey levels and reconstruct single-view colon surfaces at 7% (2.8 mm) mean depth error—in simulation; their in-vivo reconstruction has arbitrary scale, since absolute scale there requires known albedo and camera auto-gain. The illumination-decline line inherits the same ambiguity: LightDepth [15] reports 3.7 mm to 4.4 mm

MAE on the C3VD colon phantom *after median alignment to ground-truth scale*; LightNeuS [16] reaches 2.80 mm MAE with twenty views and ground-truth-registered metric poses, degrading to 8.23 mm when camera motion falls below 1 cm; and EndoMetric [17] recovers true metric scale from a ~ 3 mm light-camera baseline, at $\sim 1\%$ scale error within 8 mm of the mucosa but 5% by 20 mm (in simulation), its observability decaying with the baseline-to-distance ratio. The present work pushes the same physics to a harder operating point and a stronger evaluation: a 6.05 mm baseline at 55 mm (ratio 0.11), four sequenced lights that overdetermine the per-pixel solve in place of learning or constant-albedo assumptions, and rigid, textured bone scored against CT fiducials rather than simulation or scale-aligned metrics.

C. Surgical Surface Reconstruction and Registration

The accuracy bar for bone-surface registration is set by systems with far more favourable sensing geometry, with the spine—the target of our evaluation—the most quantified case. A tracked stylus sampling ~ 20 lamina points registers a dry-bone lumbar phantom to 0.96 ± 0.24 mm RMS [30]; structured-light optical topographic imaging acquires $>250k$ points and registers in ~ 5 s, with clinical pedicle-screw errors of ~ 1.2 mm median [31]; stereovision serves the same role in open exposures [32], with surface-based spine registration analysed as early as Herring et al. [2], and accuracy reported as fiducial and target registration error after Fitzpatrick et al. [33]. Deployed 3-D navigation delivers 1.2–1.75 mm mean translational accuracy in vivo [34]—but these are open-field or line-of-sight systems, and their static figures erode in use: vertebral displacement from surgical manipulation averages 1.6 mm, respiration displaces vertebrae by 2.0 mm on average (most in the lower thoracic spine), and error grows by ≥ 4 mm five levels from the reference frame [35]. Rigid alignments come from point correspondences [36], [37] or ICP [38]; shape-from-shading has been registered to CT in bronchoscopy via pq-space [39], and robotic spine systems have explored ultrasound as the registration modality [40]. Each of these opens the field to a projector or stereo head, adds a transducer, or relies on tracked hardware; we keep the single near-coaxial scope already in the corridor and move the difficulty into the photometric solve. Consistent with surgical practice, every vertebra is treated as its own rigid body [2].

III. HARDWARE AND EXPERIMENTAL SETUP

A. Probe Hardware

The probe is an 18 mm-OD rigid endoscope—the diameter class of tubular-retractor MIS access [1]—with a monocular camera (native 640×480 , diagonal field of view $\leq 90^\circ$) in a central $\varnothing 9.7$ mm bore and four white LEDs in tip pockets at azimuths 0° , 90° , 180° , 270° around the lens. The CAD pocket-centre radius is 5.09 mm; the as-built emitter radius, measured with calipers, is $a_r = 6.05$ mm, and the measured value is what enters the model. The emitters additionally exhibit a ~ 2 mm axial spread along the probe axis, an as-built deviation that the ideal-ring model of Sec. IV does

not capture and to which the error analysis returns. The LEDs are sequenced one at a time under constant-current DC drive (PWM dimming beats against the rolling shutter and masquerades as shading), with exposure and gain locked for the whole session.

B. Phantom, Fiducials, and Capture Protocol

The target is a rigid lumbar phantom (Sawbones 1324-56, L1–sacrum, solid foam), unmodified, instrumented with four retroreflective spherical fiducials per vertebra (level 7: three usable) and CT-scanned with the fiducials in place. Each fiducial is visible both in CT (a surface crater) and in the images (a specular dot), providing the 2-D/3-D correspondences that serve as ground truth for evaluation. Each capture records a dark frame plus four single-LED frames from a fixed pose, flushing the camera’s frame buffer after every LED switch so the four images cannot be misassigned.

C. Calibration and Radiometric Preprocessing

Intrinsics and distortion come from a ChArUco calibration (121 views, 0.11 px reprojection); the working focal length $f \approx 860$ px is additionally validated by four-point PnP reprojection (0.42 px residual) using SQPNP [41]. The photometric solve treats the mildly distorted pixel grid as pinhole; the distortion model is applied when projecting for pose validation. Frames are converted to linear luminance in four steps: dark subtraction, sRGB→linear decoding, Rec. 601 luminance, and specular inpainting of bright low-saturation pixels (the retroreflective fiducials violate Lambert). Because frames are captured sequentially, each is divided by its median to cancel per-frame exposure drift—a spatially constant correction that leaves the shading field intact; the residual per-LED power difference is absorbed by per-LED gains estimated inside the objective (Sec. IV-C).

D. Near-Coaxial Conditioning

At working distance b each LED subtends $\alpha = \arctan(a_r/b) \approx 6^\circ$, so a far-field light direction is $\mathbf{L}_s = [\sin \alpha \cos \phi_s, \sin \alpha \sin \phi_s, \cos \alpha]^\top$. Stacking the four ring azimuths, $\mathbf{L}^\top \mathbf{L} = \text{diag}(2 \sin^2 \alpha, 2 \sin^2 \alpha, 4 \cos^2 \alpha)$, so the condition number of the light matrix is exactly $\sqrt{2} \cot \alpha \approx \sqrt{2} b/a_r$ (≈ 13 at 55 mm): image noise is amplified into the tangential normal components by $\sim 1/\sin \alpha$, recovered relief is compressed, and errors in the assumed light geometry are magnified by the same factor. This is the regime the method is designed for.

E. Wiring and Aim Identification

The map from frame index to ring azimuth must be established before any reconstruction. Scoring all $4!$ assignments by the correlation between reconstructed depth at the fiducials and the fiducial depths of two levels identifies the physical wiring (LED1→ 90° , LED2→ 270° , LED3→ 180° , LED4→ 0°) at $r = +0.97/+0.99$, while the diametrically flipped labelling *inverts* the surface ($r = -0.97/-0.99$): a mislabelling is not a perturbation but a sign flip of the recovered geometry, a

silent failure mode for any multi-LED rig. An independent cue corroborates the assignment: dividing each single-LED image by the four-LED mean cancels albedo and shape, and the brightness-weighted centroid of the ratio image marks where that LED’s *beam* lands. Unanimously across all five captures, the vertical pair’s beams land on the side opposite their ring positions: those LEDs are aimed convergently, crossing the optical axis before the working distance. Beam aim redistributes intensity but cannot flip the side light arrives from, so incidence—and hence the wiring—is set by LED position; the aim finding instead shows that a parallel-aim emission model is only an approximation on this rig (Sec. IV-E). A one-time bench check (powering each LED individually) would confirm the assignment directly.

IV. METRIC NEAR-FIELD RECONSTRUCTION

A. Near-Field Model and Free-Normal Solve

In the camera frame (+ x right, + y up, + z toward the camera) a pixel (u, v) at depth D back-projects to $\mathbf{X} = D[(u - c_x)/f_x, -(v - c_y)/f_y, -1]^\top$, and LED s sits at $\mathbf{P}_s = [a_r \cos \phi_s, a_r \sin \phi_s, 0]^\top$. With incident direction $\mathbf{l}_s = (\mathbf{P}_s - \mathbf{X})/\|\mathbf{P}_s - \mathbf{X}\|$ and attenuation

$$w_s = \frac{\cos^\mu \theta_s}{\|\mathbf{P}_s - \mathbf{X}\|^2}, \quad \cos \theta_s = \frac{D}{\|\mathbf{P}_s - \mathbf{X}\|}, \quad (1)$$

the point-source model $I_s = \rho(\mathbf{n} \cdot \mathbf{l}_s) w_s$ —with the $\cos^\mu$ emitter anisotropy standard for LEDs [42], aimed along the optical axis—linearises by moving the attenuation to the observation, $m_s := I_s/w_s = \mathbf{g} \cdot \mathbf{l}_s$ with $\mathbf{g} = \rho \mathbf{n}$, giving a per-pixel 4×3 least-squares system

$$\hat{\mathbf{g}} = (\mathbf{\Lambda}^\top \mathbf{\Lambda} + \epsilon \mathbf{I})^{-1} \mathbf{\Lambda}^\top \mathbf{m}, \quad \mathbf{\Lambda} = [\mathbf{l}_1 \ \mathbf{l}_2 \ \mathbf{l}_3 \ \mathbf{l}_4]^\top. \quad (2)$$

The normals $\hat{\mathbf{g}}/\|\hat{\mathbf{g}}\|$ are integrated by Frankot–Chellappa [20] (even reflection, zero DC) into a *relative* relief $\tilde{z}(u, v)$, median-centred (a 1–99-percentile clip is applied only to the downsampled copy used inside the search objective; the full-resolution relief that forms the returned surface is unclipped). Since $\mathbf{\Lambda}$ and w_s depend on D , the solve is iterated with the current depth estimate. Up to here the pipeline is standard near-field photometric stereo; the question is what metric depth to attach to $\tilde{z}$.

B. A Two-Parameter Metric Depth Family

We parameterise the metric surface by a working distance b (mm) and a dimensionless relief gain s ,

$$D_{b,s}(u, v) = b \left(1 + s \frac{\tilde{z}(u, v)}{f} \right), \quad (3)$$

where dividing by the focal length f of the grid over which $\tilde{z}$ was integrated makes the family resolution-invariant and puts s on a natural scale: $s = 1$ is the perspective-consistent gain, so s reads out directly how much the near-coaxial solve compresses or inflates relief. Under far-field lighting a family like (3) would be unobservable—relief gain is the flattening direction of the generalised bas-relief ambiguity [22]—and it is exactly the near-field terms that break it [26]. (b, s) are also

precisely the two degrees of freedom that anchored near-field pipelines take from an external depth reference; here they are estimated from the images.

C. The Tied-Normal Photometric Objective

With four lights and three per-pixel unknowns, (2) is over-determined, so a residual survives any inconsistency between the four images and the assumed geometry. Where the normals in that residual come from is decisive. Solved freely per pixel, they tilt to compensate a wrong (b, s) and the objective goes flat. Candidates are therefore evaluated with normals *tied* to the surface: from $D_{b,s}$ we form the 3-D point map $\mathbf{X}(u, v)$, take finite-difference tangents $\partial_u \mathbf{X}, \partial_v \mathbf{X}$, and set $\mathbf{n} = \pm \partial_u \mathbf{X} \times \partial_v \mathbf{X}$ oriented toward the camera. The model shading is $\hat{m}_s = (\mathbf{n} \cdot \mathbf{l}_s) w_s$ clamped at zero, and the only remaining freedoms are physically nuisance ones: a per-pixel albedo ρ , one gain γ_s per LED, and the rig-level emission exponent μ . Albedo and gains have closed-form least-squares updates, alternated twice:

$$E(b, s; \mu) = \left\langle \frac{\sum_s \delta_s |I_s - \rho \gamma_s \hat{m}_s|}{\sum_s \delta_s I_s} \right\rangle_{\text{mask}}, \quad \delta_s = [\hat{m}_s > 0], \quad (4)$$

so attached shadows are dropped from both sums. Albedo absorbs texture and common-mode vignetting; the γ_s absorb LED power differences; μ absorbs the emission profile; what remains is geometry. A wrong b mis-scales the inverse-square and emission terms across the field; a wrong s mis-tilts every normal; both raise E . Absolute scale enters because a_r in (1) is a measured hardware length: multiplying (b, D) by a constant does not leave the imaging geometry invariant. This is the observability that far-field photometric stereo lacks—the attenuation term is also what makes near-source shape-from-shading well-posed [23]—and that a larger light baseline strengthens [17].

D. Search: Focus-Band Prior, Multi-Start, Keep-Best

E is cheap on $4 \times$ -downsampled images, but it is shallow and multi-modal in (b, μ) on this near-coaxial rig: a joint search can fall into a false “close-and-flat” basin. Three elements regularise the minimisation. First, b is constrained to the lens depth-of-field band (45 mm to 70 mm), a bench-measured property of the fixed-focus optics—captures are only sharp in this range—which excludes the false basin; focus is itself a classical coarse depth cue [43], exploited as such in endoscopes [44], and here it contributes exactly one interval, not a depth map. Second, μ is not free per capture: it is hardware, fixed to the rig-level value μ^* (Sec. IV-E). Third, the alternation between the free-normal relief $\tilde{z}$ and the (b, s) search (Alg. 1) is not monotone— $\tilde{z}$ moves under the search—so the iterate with the lowest E is kept, from each of three starts spread across the focus band, and the best over starts is returned. A coarse (b, s) grid (s sampled over 0.25–3 within the bound $[0.1, 5]$; b steps of 5 mm within ± 20 mm of the current iterate) seeds a Nelder–Mead refinement in $(b, \log s)$, and depth is floored at 2 mm so no candidate surface passes behind the lens.

Algorithm 1 Metric near-field reconstruction

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1: for  $b_0 \in \{48, 57, 66\}$  mm do
2:    $D \leftarrow$  flat plane at  $b_0$ 
3:   for  $k = 1$  to 6 do
4:     free-normal solve (2) at  $D$ ; integrate  $\rightarrow \tilde{z}$ 
5:      $(b, s) \leftarrow \arg \min E(b, s; \mu^*)$  (4), grid + refine
6:      $D \leftarrow b(1 + s\tilde{z}/f)$ ; record  $(E, b, s, \tilde{z})$ 
7:   end for
8: end for
9: return lowest- $E$  record over all starts and iterations

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E. Rig-Level Emission Exponent

The LEDs’ angular fall-off $\cos^\mu \theta$ is a property of the rig, not the scene, so all captures share one μ^* . The photometric residual alone cannot select it: a cross-capture vote—each candidate μ scored per capture by the best residual achievable under Alg. 1 with μ pinned, per-capture normalised, then summed—varies by less than 6% over $\mu \in [0, 10]$. Final accuracy, in contrast, depends on μ strongly. μ^* is therefore selected end-to-end on two levels: candidate values are scored against the fiducial ground truth of levels 4–5, both of which prefer $\mu \approx 8$, and $\mu^* = 8$ is held fixed for all captures. Level 6, excluded from this selection, reconstructs at 2.78 mm with the transferred constant, against 8.35 mm at the vote’s weak minimum ($\mu = 2$)—the same setting also degrades level 5 to 5.52 mm. This insensitivity of the shading objective paired with strong sensitivity of the final accuracy indicates that μ must come from calibration rather than from the solve; a one-time bench white-target measurement of the per-LED profile [12], [45], [46] is the standard procedure, and the convergent-aim finding (Sec. III-E) argues that it should include per-LED aim.

F. Surface Extraction

A pixel is kept if bright enough, photometrically consistent (final free-normal residual < 0.22), within a robust depth band about the median of the reconstruction, and not a spike; the border is cropped (integration ramps), speckle is removed by morphological opening, components smaller than $\max(300 \text{ px}, 15\% \text{ of the largest})$ are dropped, the mask is eroded once, and the surviving depth is mask-aware smoothed and triangulated on the image grid, emitting quads only where the four corners span $< 3 \text{ mm}$.

V. EXPERIMENTS

A. Protocol and Metrics

Five vertebral levels of the instrumented phantom of Sec. III-B were captured at working distances of 51 mm to 60 mm. Reconstruction uses only the four images and the rig parameters of Secs. III–IV; the fiducial correspondences serve as ground truth. Each correspondence pairs the CT fiducial point (marker crater) with its glint-centroid pixel in the images, whose depth is read from the raw reconstruction as the median over a 14-px-radius disk, before trimming and smoothing—identically for every method compared. Three metrics are

TABLE I

PER-LEVEL RESULTS. b : RECOVERED WORKING DISTANCE (TRUE VALUE IN PARENTHESES); s : RELIEF GAIN; SCALE: SIMILARITY (UMEYAMA); FRE: 3-D FIDUCIAL RMS, RIGID 6-DOF ALIGNMENT FOR OUR COLUMN. THE ORACLE AND FAR-FIELD VARIANTS USE THE DEPTH-ANCHORED PROTOCOL OF SEC. V-A.

Level	Ours				depth-anchored	
	b (mm)	s	scale	FRE (mm)	oracle	far-field
4	45.1 (60.0)	0.74	0.85	2.68	1.72	1.54
5	45.6 (55.4)	1.00	1.27	3.07	1.32	1.22
6	52.3 (58.9)	1.00	1.05	2.78	1.74	6.57
7 [†]	69.9 (58.6)	1.86	0.77	6.02	9.72	11.14
8	unstable (mask collapse)				unstable	5.66

[†]three fiducials: minimal for a 6-DOF alignment, degenerate for PnP.

reported. (i) **Registration-free scale**: ratios of pairwise fiducial distances, reconstruction over CT. (ii) **Rigid-only FRE**: the fiducial RMS after the best 6-DOF Kabsch alignment [36]—no similarity scaling, so scale error is penalised, unlike the 7-DOF fits common in registration studies. (iii) **Similarity scale**: the Umeyama scale [37] c of the fit $\text{CT} \approx c R \text{rec} + t$, which is 1 for a metric-true surface ($c < 1$: reconstruction too large). We report fiducial—not target—registration error and note the two are distinct [33]. Three reference methods run under the identical registration protocol: the classical far-field solve; direct inverse-square shading-to-range; and an *oracle-anchored* variant of the near-field solve that is given the true working distance and an affine depth fit to the fiducials at every iteration ($\mu = 0$, four iterations, fiducial-band trim)—an upper bound on what this rig achieves when an external depth reference is available.

B. Metric Reconstruction Accuracy

Table I and Fig. 2 summarise the results. On all three well-conditioned four-fiducial levels the surfaces align rigidly to CT at 2.68 mm to 3.07 mm RMS, with similarity scales of 0.85–1.27; the registration-free per-level pairwise-distance ratios deviate from unity by 10% to 21%, and recovered working distances lie within the focus band (levels 4–5 at or near its lower edge; Sec. V-E) but 6.6 mm to 15 mm from the true values. The oracle-anchored variant reaches 1.32 mm to 1.74 mm on identical data: the gap is a factor ≈ 2 , attributable mostly to the residual scale error, which a 6-DOF alignment cannot absorb. Published accuracy requirements for image-guided pedicle-screw placement are of order 1 mm translational at the tightest levels [47]; Sec. V-F places the result against that bar and the wider literature.

C. What the Residual Does and Does Not Select

Fig. 3 shows the flat-plane residual landscape $E(b, 1; \mu)$ and a convergence trace. The landscape is informative but shallow, with a (b, μ) trade-off valley—fixing μ^* removes this degeneracy, and the focus band removes the false close-and-flat basin that remains. On level 4 that removal is literal: the descent is stopped by the band’s lower edge rather than by an

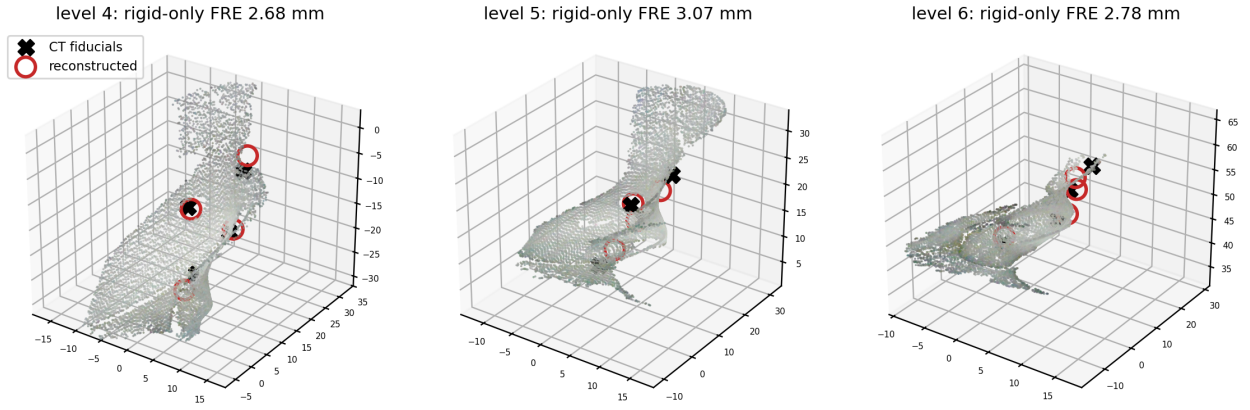


Fig. 2. Reconstructed surfaces of the three well-conditioned levels in CT coordinates after a rigid 6-DOF alignment (no scale correction). Open circles: reconstructed fiducials; crosses: ground truth. The residual mismatch is dominated by the scale error of the solve (Sec. V-B), which the 6-DOF alignment does not absorb.

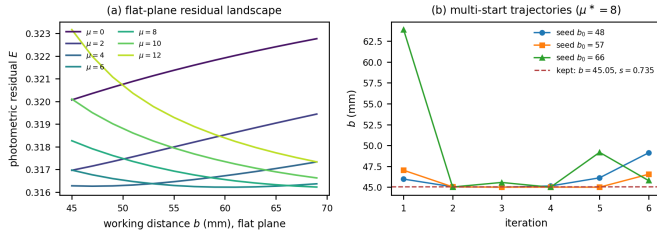


Fig. 3. Level 4. (a) Flat-plane photometric residual vs. working distance b for each candidate emission exponent μ : the total variation across the focus band is $< 2.5\%$ —informative but shallow, and nearly μ -degenerate, motivating the focus-band prior and the pinned rig-level μ^* . (b) Working-distance traces of Alg. 1 from the three seeds; all descend to $b \approx 45$ mm, and the photometrically best iterate (dashed) is kept.

interior minimum (Fig. 3b), so there the prior, not the residual, sets b ; level 5 stops within 0.55 mm of the same edge.

D. Baselines

Far-field. The far-field solve integrates its geometric mis-fit into a low-frequency bowl (Fig. 4); even with depth anchoring it reaches millimetre FRE only where the surface happens to be benign (levels 4–5) and fails structurally on level 6 (6.57 mm), which the near-field model repairs (1.74 mm). **Inverse-square.** Estimating range directly from brightness (range $\propto I^{-1/2}$) with identical anchoring fails on every level—6.4 mm to 9.6 mm FRE and grossly inflated relief—because it conflates albedo with range on textured bone; the albedo-invariant normal-based solve is not optional.

E. Failure Cases

Level 7 has three fiducials: the rigid alignment is minimally constrained, the scale check rests on three pairwise distances, and the recovered b rails at the top of the focus band; FRE degrades to 6.02 mm. The anchored protocol is itself degenerate with three fiducials (P3P admits up to four poses), so its 9.72 mm is not a meaningful comparator. *Level 8* collapses under both protocols—the near-field confidence mask empties

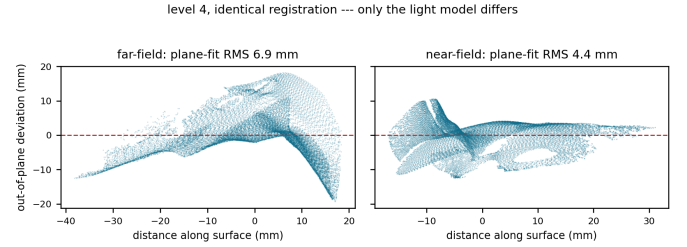


Fig. 4. Level 4, identical data and registration, only the light model differs: out-of-plane profiles of the reconstructed surface. The far-field solve (left) integrates a systematic bowl (plane-fit RMS 6.9 mm); the near-field solve (right) flattens it (4.4 mm; the remainder is genuine bone relief).

while the far-field solve on the same frames stays stable—an unresolved solver-robustness issue, not a distance effect (all captures lie at 51 mm to 60 mm). Separately, the recovered b sits at or within 0.6 mm of a band edge on levels 4, 5, and 7 (0.05, 0.55, and 0.11 mm respectively), indicating that the b -gradient of E is weak at this α ; band-edge convergence is therefore a useful per-capture self-diagnostic.

F. Accuracy in Context

Table II situates the result; metrics and conditions differ across rows. Three readings emerge. First, 2.7 mm to 3.1 mm from a single pose of a four-LED scope is inside the accuracy class of the best calibrated near-field photometric stereo on a lab rig with 52 LEDs at 10–30 cm (3.2 mm mean depth error [24]) and of multi-view neural reconstruction given ground-truth-registered poses (2.80 mm [16])—obtained here with no pose input, no parallax, no learning, and scale from a 6.05 mm baseline at 55 mm. The 10% to 21% scale errors are likewise in class: EndoMetric, the comparable illumination-baseline system evaluated on real scenes, deviates 13% from the endoscopist’s estimate on real polyps at a more favourable baseline-to-distance ratio [17]. Second, it does not yet match contact or structured-light registration (0.96 mm to 1.2 mm [30], [31]) or deployed navigation (1.2–

TABLE II
REPORTED ACCURACY OF RELATED SYSTEMS. METRICS AND
CONDITIONS DIFFER (RMS VS. MAE; PHANTOM VS. CLINICAL); THE
TABLE SITUATES, IT DOES NOT RANK.

Method	Evaluation	Error (mm)
Ours (1 pose, 4 LEDs, 55 mm) + oracle depth anchor	phantom, CT fid. phantom, CT fid.	2.68–3.07 1.32–1.74
Tracked stylus, 20 pts [30]	dry-bone phantom	0.96 RMS
Structured light [31]	clinical screws	~1.2 med.
3-D navigation [34]	clinical screws	1.2–1.75
LightNeuS, 20 views + poses [16]	colon phantom	2.80 MAE
Single-view near-light [13]	simulation	2.8 (7%)
LightDepth, scale-aligned [15]	colon phantom	3.7–4.4 MAE
Near-field PS, 52 LEDs [24]	lab objects	3.17 depth

1.75 mm [34])—though the structured-light system’s clinical 95th-percentile errors reach 3.4 mm to 4.3 mm [31]; the oracle-anchored bound (1.32 mm to 1.74 mm) shows that the photometric solve supports that class when depth is externally anchored; recovering it without the anchor is what bench LED calibration targets. Third, the operating point differs in kind: those systems need an open exposure or line of sight, which the corridors that motivate this work—tubular retractors, arthroscopic portals, endonasal approaches—do not offer; and navigation accuracy silently erodes intraoperatively—manipulation, respiration, and distance from the reference frame each contribute millimetres [35]—while a sensor inside the corridor could re-register on demand without re-imaging. That, not raw accuracy, is the niche this class of system targets.

G. Discussion and Limitations

The dominant residual is the assumed LED geometry: positions on an ideal coplanar ring and a symmetric $\cos^\mu$ emission, both contradicted in detail by the as-built hardware—the measured ~2 mm axial emitter spread (Sec. III-A) and the convergent aim (Sec. III-E). Bench calibration of per-LED position, aim, and profile [12], [27], [45] is the clear next step and should tighten both scale and b . The focus band is a weak prior in principle, but it is a lens property, measurable once on the bench. Validation covers a single anatomy—the posterior elements of one rigid lumbar foam phantom—whose reflectance differs from wet bone (subsurface scatter, blood, and instrument clutter are untested); articular and skull-base bone differ again in reflectance, moisture, and working distance, so extension to those corridors remains to be demonstrated. Per-level fiducial counts limit the statistical strength of each FRE. Runtime (minutes per capture, unoptimised Python) is compatible with a registration step, not with live video.

VI. CONCLUSIONS

A near-coaxial four-LED endoscope can measure exposed bone metrically from a single pose: tying candidate normals to a two-parameter depth family turns the redundancy of the four-light solve into a selector for working distance and relief gain, with the LED ring baseline providing absolute scale and a bench focus band as the only prior. On a CT-instrumented lumbar spine phantom the method reaches 2.7 mm to 3.1 mm

rigid-only fiducial RMS on every well-conditioned level—within a factor 1.6–2.3 of an oracle-anchored bound—where far-field photometric stereo suffers a structural bowl and inverse-square ranging fails outright. Bench LED calibration to close the oracle gap, a working-distance sweep, and additional specimens are the direct next steps toward in-corridor registration—per vertebra in the spine, and beyond it wherever bone meets a narrow approach—from the scope already in the field, with no added radiation.

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